Software & Perception

WORKSHOP

2026

Duration	8 October 2026 15 November 2026 (six calendar weeks, 8 core sessions), preceded by an online pre-workshop session on 6 October 2026. Saturday and Sunday sessions run through to 14 15 November.
Schedule	Core sessions: Tuesday and Thursday (Session 8 on Friday, 13 November) Industry expert sessions: Saturdays (except 17 Oct and 14 Nov) Fundamentals sessions: Saturday 14 and Sunday 15 November Weekly catch-up: every Sunday, 6:00 PM (until 8 November)
Timing	Core sessions: 5:10 PM 6:50 PM (1 h 40 min) Saturday industry sessions: half day (timing to be notified) Fundamentals sessions (14 and 15 Nov): timing to be notified Sunday catch-up: 6:00 PM
Holiday	No sessions on Saturday, 17 October 2026 and Tuesday, 20 October 2026 (Vijayadashami)
Subjects	System Design, Computer Vision, Machine Learning, Agentic AI, ROS 2 and Software Engineering Fundamentals; closing sessions on Automobiles, EVs and Autonomous Vehicles
Format	8 sessions, each with its own mini project, and two larger projects: Project 1 after the Computer Vision session (15 Oct) and Project 2 after the workshop. Industry expert sessions on Saturdays, weekly catch-ups every Sunday, fundamentals sessions on 14 15 Nov, and guest sessions by seniors and alumni run alongside.
Support	Handbooks, guides and all required material provided to every participant
Certificate	**Attendance will be monitored. Certificates will be given only to participants who meet the attendance requirement.**

Workshop Objectives

- Learn to break a complex vehicle into subsystems and show it through clear block diagrams.
- See how a camera perceives the world and how a machine learns from data, through practical vision and prediction projects.
- Understand how the software parts of an autonomous vehicle talk to each other using ROS 2.
- Explore how modern AI agents reason, use tools and make decisions, and apply them to engineering problems.
- Hear from industry experts on how real vehicles are designed, built and tested.
- Build a foundation in automobiles, EVs and autonomous vehicles in the closing weekend sessions.
- Learn by building: a mini project in every session, plus two larger projects.

1. Colour Legend

- **Pre-Workshop Session**
- **System Design**
- **Computer Vision**
- **Machine Learning**
- **ROS**
- **Agentic AI**
- **Guest Session**
- **Industry Expert Session**
- **Fundamentals Session**
- **Weekly Catch-up**
- **No Session (Holiday)**

LAPTOP	TIMINGS	DAYS	STARTS
Mandatory for every session	**5:10 PM 6:50 PM**	**Tue & Thu (+ Fri 13 Nov) Sat (industry) Sun 6 PM (catch-up)**	**Thu 8 Oct 2026 (online pre-workshop: Tue 6 Oct)**

Participant Guidelines

- **Bring your laptop.** A laptop is mandatory for every session. Carry your charger as well, and keep the laptop charged before you arrive.
- **Be on time.** Sessions run from 5:10 PM to 6:50 PM. Please be seated and ready by 5:10 PM so that no content is missed.
- **Finish the setup.** Complete the prerequisites and software setup covered in the pre-workshop session (Tue 6 Oct) before the first session on Thu 8 Oct.
- **Attendance is monitored.** Attendance is recorded for every session, and certificates will be given only to participants who meet the attendance requirement.
- **Do the mini projects and projects.** Every session has its own mini project. Project 1 is given after the Computer Vision session (Thu 15 Oct) and Project 2 after the workshop. Details and deadlines will be shared in due course.
- **Attend the Saturday industry sessions.** These half-day sessions bring in experts from industry; timings will be shared before each one.
- **Missed a class?** Use the Sunday catch-up sessions at 6:00 PM to revisit the content and keep pace with the syllabus.
- **Keep the material handy.** Use the handbooks, guides and study material provided; they are your reference during and after the workshop.
- **Ask questions.** Raise doubts with the workshop team at any point. Suggestions to improve the workshop are always welcome.
- **Check for updates.** Industry session speakers and timings, fundamentals session timings, and guest session dates will be communicated in due course; the schedule may change with or without prior notice.

2. Timetable

Core sessions: 5:10 PM 6:50 PM Tuesday and Thursday (Session 8: Friday 13 Nov) ***Saturday: industry expert session (half day, timing to be notified) Sunday 6:00 PM: weekly catch-up 14 15 Nov: fundamentals sessions (timing to be notified)***

No.	Date	Subject	Mini project / Project	Presented by
Week 1 5 11 Oct				
Pre	**6 Oct 2026** *Tuesday*	**Pre-Workshop Session**		**All team members**
1	**8 Oct 2026** *Thursday*	**System Design**	Mini Project 1 (System Design)	**Rithvin, Preethika**
Ind	**10 Oct 2026** *Saturday*	**Industry Expert Session** Half day timing to be notified		Industry expert (to be notified)
CU	**11 Oct 2026** *Sunday*	**Weekly Catch-up** 6:00 PM		Workshop team
Week 2 12 18 Oct				
2	**13 Oct 2026** *Tuesday*	**System Design**	Mini Project 2 (System Design)	**Rithvin, Preethika**
3	**15 Oct 2026** *Thursday*	**Computer Vision**	Mini Project 3 (Computer Vision) **+ Project 1** assigned after this session	**Mr. Mahavishnu V C**
	17 Oct 2026 *Saturday*	**No session (Vijayadashami)**		
CU	**18 Oct 2026** *Sunday*	**Weekly Catch-up** 6:00 PM		Workshop team
Week 3 19 25 Oct				
	20 Oct 2026 *Tuesday*	**No session (Vijayadashami)**		
4	**22 Oct 2026** *Thursday*	**Machine Learning** Introduction to ML and Unsupervised Training	Mini Project 4 (ML)	**Jeevan**
Ind	**24 Oct 2026** *Saturday*	**Industry Expert Session** Automotive Functional Safety half day, timing to be notified		Industry expert (to be notified)
CU	**25 Oct 2026** *Sunday*	**Weekly Catch-up** 6:00 PM		Workshop team
Week 4 26 Oct 1 Nov				
5	**27 Oct 2026** *Tuesday*	**Agentic AI** Introduction to Agentic AI 1	Mini Project 5 (Agentic AI)	**Arya, Josana**
Ind	**31 Oct 2026** *Saturday*	**Industry Expert Session** Half day timing to be notified		Industry expert (to be notified)
CU	**1 Nov 2026** *Sunday*	**Weekly Catch-up** 6:00 PM		Workshop team
Week 5 2 8 Nov				
6	**3 Nov 2026** *Tuesday*	**Agentic AI** Agentic AI Basics 2	Mini Project 6 (Agentic AI)	**Arya, Josana**
Ind	**7 Nov 2026** *Saturday*	**Industry Expert Session** Half day timing to be notified		Industry expert (to be notified)
CU	**8 Nov 2026** *Sunday*	**Weekly Catch-up** 6:00 PM		Workshop team
Week 6 9 15 Nov				
7	**10 Nov 2026** *Tuesday*	**Machine Learning** Supervised Learning: From the Basics	Mini Project 7 (ML)	**Jeevan**

No.	Date	Subject	Mini project / Project	Presented by
8	**13 Nov 2026** *Friday*	**ROS 2** ROS 2 and Software Engineering Fundamentals	Mini Project 8 (ROS) **Project 2** follows after the workshop	**Ratheeshwar**
Fun	**14 Nov 2026** *Saturday*	**Fundamentals of Automobiles and EVs** Timing to be notified		To be notified
Fun	**15 Nov 2026** *Sunday*	**Fundamentals of Autonomous Vehicles** Timing to be notified		To be notified
After the workshop				
P2	**After 15 Nov** *dates to be notified*	**Project 2**	**Project 2** done after the workshop	Workshop team
To be notified				
Ind	**Saturday** *date to be notified*	**Industry Expert Session** Vehicle Dynamics and Architecture		**Mr. Senthil Vel M (VIT), Mr. Vimal Kumar (MEI)**
	To be *notified*	**Guest Sessions I & II** Powertrain & Embedded/Electrical; Mechanical fundamentals		Senior students and alumni

Ind = Industry expert session CU = Weekly catch-up Fun = Fundamentals session P2 = Project 2. The Vehicle Dynamics and Architecture session is held on one of the Saturdays listed; its date will be notified. Saturday sessions on 31 Oct and 7 Nov and the Sunday catch-up on 8 Nov are retained from the earlier plan so that every Saturday and Sunday of the workshop calendar is covered. There are no core sessions on Thursday 29 Oct or Thursday 5 Nov.

3. Session-by-Session Guide

PRE-WORKSHOP ONLINE

Pre-Workshop Session: Set Up Once, Build All Month

Date	Tuesday, 6 Oct 2026	Time	Online session
Subject	Pre-Workshop Session	Presented by	**All team members**

A short online session so that nobody loses time to installation on day one. We walk through the prerequisites, get every laptop ready, and give you a map of the weeks ahead.

What you will learn

- A working development environment for every session
- A clear picture of the workshop structure, topics, and weekend sessions
- Accounts and keys you'll need later, set up in advance
- A place to ask for help if anything fails to install

Topics covered

- Prerequisites and requirements
- Setup checklist: Ubuntu 24.04, ROS 2 Jazzy, Python 3 and libraries, VS Code, Git and GitHub, Google Colab, LLM API key or Ollama
- Overview of the workshop
- Help with installation and troubleshooting

Tools & tech stack

Ubuntu 24.04 / WSL2 ROS 2 Jazzy Python 3 VS Code Git Colab

SESSION 1 SYSTEM DESIGN

System Design I: Subsystem Thinking

Date	Thursday, 8 Oct 2026	Time	5:10 PM 6:50 PM
Subject	System Design	Presented by	**Rithvin, Preethika**
Mini project	**Mini Project 1 (System Design)**		

Before a single line of code runs on the car, someone has to decide what talks to what. This session teaches you to break a full autonomous ATV into clean, buildable subsystems.

What you will learn

- Decompose a complex vehicle into inputs, processing units and outputs across mechanical, electrical and software domains
- Explain the Sense Plan Act loop that every autonomous vehicle is built around
- Identify the sensors, compute and actuators of an ATV and where each belongs

Topics covered

01 **Subsystem Thinking**
Breaking the ATV into Perception, Localisation, Planning, Control, Actuation and Power. Mapping each to its inputs, processing unit and outputs across all three domains.

02 **The Sense Plan Act Loop**
How cameras, IMU, GPS and wheel encoders feed a compute unit that decides steering, throttle and brake commands, and why loop timing matters.

03 **A Tour of an Autonomous ATV**
Walking through a real architecture end to end to see how the subsystems fit together.

Tools & tech stack

draw.io / diagrams.net Excalidraw Pen & paper

Before you arrive

None. A laptop with a browser is enough.

Why it matters on the car Every other session in this workshop plugs into the system you map today: CV and ML live in the Perception block, ROS is the wiring between blocks, and Agentic AI sits on top as a tool for engineers.

SESSION 2 SYSTEM DESIGN

System Design II: Block Diagramming

Date	Tuesday, 13 Oct 2026	Time	5:10 PM 6:50 PM
Subject	System Design	Presented by	**Rithvin, Preethika**
Mini project	**Mini Project 2 (System Design)**		

Turn the subsystems from Session 1 into diagrams the way real engineering teams draw them, with clean interfaces and a safe design.

What you will learn

- Read and draw block diagrams using standard conventions for data flow, power distribution and physical linkages
- Define clean interfaces between subsystems (signal type, voltage, data rate, connector) so teams can work in parallel

- Spot single points of failure and place an emergency-stop path in a design

Topics covered

01 **Block Diagramming Conventions**
Arrow styles for data vs power vs mechanical linkage, colour-coding by domain, labelling interfaces (e.g. 12 V, CAN @ 500 kbps, USB 3.0), and keeping diagrams readable.

02 **Interfaces & Contracts**
Why defining the boundary between two subsystems first lets mechanical, electrical and software teams build at the same time.

03 **Safety by Design**
Failure modes, redundancy and the hardware e-stop: what must still work when the software crashes.

Tools & tech stack

draw.io / diagrams.net Excalidraw Pen & paper

Before you arrive

Session 1 notes reviewed; a laptop with a browser.

Why it matters on the car A clear diagram with defined interfaces is what lets a whole team build one vehicle in parallel without stepping on each other.

SESSION 3 COMPUTER VISION

Computer Vision: Teaching the Car to See

Date	Thursday, 15 Oct 2026	Time	5:10 PM 6:50 PM
Subject	Computer Vision	Presented by	**Mr. Mahavishnu V C**
Mini project	**Mini Project 3 (Computer Vision)**	Project	**Project 1 is assigned after this session**

A camera gives the car millions of numbers per second. This session shows you how to turn those numbers into something useful; by the end you will be able to find and track objects in live video.

What you will learn

- Explain how a digital image is stored: pixels, resolution, channels and bit depth
- Load, display, resize and crop images and video frames with OpenCV
- Reduce camera noise with Gaussian blur and explain what the kernel size does
- Convert between BGR and HSV and isolate an object by its colour
- Find contours, draw bounding boxes and compute an object's centre point

Topics covered

01 **Pixels, Resolution & Channels**
An image as a NumPy array of shape (height, width, channels). Grayscale vs colour, 8-bit values 0 255, and why OpenCV stores colour as BGR, not RGB.

02 **Image Manipulation Basics**
Resizing (and why it speeds up processing), cropping a region of interest, and Gaussian blur to reduce camera noise before detection.

03 **Colour Spaces & Thresholding**
Why HSV separates colour from brightness, and how cv2.inRange builds a binary mask that holds up better under changing sunlight than raw RGB.

04 **Morphology & Contours**
Cleaning masks with erosion and dilation, finding contours, filtering them by area, and drawing bounding boxes.

05 **Colour & Contour Tracking**
Running the full pipeline frame by frame on video to track a moving object and report its position.

Tools & tech stack

Python 3 OpenCV NumPy Jupyter / Google Colab Webcam (optional)

Before you arrive

Basic Python: variables, loops, functions. Python 3 and OpenCV installed, or a Google account for Colab.

Why it matters on the car Colour and contour pipelines are fast, need no training data, and are a common first stage on real vehicles, used for cone, lane and marker detection before any neural network is involved.

SESSION 4 MACHINE LEARNING

ML I: Introduction to ML and Unsupervised Training

Date	Thursday, 22 Oct 2026	Time	5:10 PM 6:50 PM
Subject	Machine Learning	Presented by	**Jeevan**
Mini project	**Mini Project 4 (ML)**		

Instead of hand-writing every rule, what if the car could learn from its own logs? This session takes the mystery out of machine learning, starting from real vehicle data.

What you will learn

- Tell the difference between supervised and unsupervised learning and pick the right one for a problem
- Explain features, labels, training and prediction
- Load, inspect and clean a vehicle log with pandas and plot it with Matplotlib

- Apply K-Means clustering to discover driving styles in unlabelled data

Topics covered

01 **What is Machine Learning?**
Rules written by hand vs rules learned from data. Features, labels, training and prediction.

02 **Core Paradigms**
Supervised learning (learning from labelled vehicle logs) versus unsupervised learning (grouping driving styles with no labels at all).

03 **Working with Data**
Reading CSV logs with pandas, plotting with Matplotlib, handling missing values, and why a train/test split is non-negotiable.

04 **Unsupervised Training in Practice**
K-Means clustering on speed, throttle and braking features to separate driving styles (smooth, aggressive, stop-and-go). Choosing K with the elbow method.

Tools & tech stack

Python 3 pandas scikit-learn Matplotlib Google Colab

Before you arrive

Basic Python. No prior maths beyond school-level graphs (y = mx + c) is needed.

Why it matters on the car Energy prediction, sensor calibration and anomaly detection on real vehicles all start from this workflow: clean the data, choose the right kind of learning, and test it honestly.

SESSION5 AGENTICAI

Introduction to Agentic AI 1

Date	Tuesday, 27 Oct 2026	Time	5:10 PM 6:50 PM
Subject	Agentic AI	Presented by	**Arya, Josana**
Mini project	**Mini Project 5 (Agentic AI)**		

Chatbots answer questions. Agents get things done: they observe, think, plan and act, often using tools along the way. This session builds a clear picture of what AI agents are and how they work under the hood.

What you will learn

- Explain what generative AI and an AI agent are, in plain terms and with real examples
- Tell the difference between an LLM, a chatbot and an AI agent
- Describe the agent loop: perceive reason plan act, and why it repeats
- Write clear, well-structured prompts
- Recognise where agents are useful in engineering and where a simple prompt is enough

Topics covered

01 **Introduction to Generative AI**
From text generators to systems that pursue a goal on their own.

02 **LLMs vs Chatbots vs AI Agents**
The model itself, a conversational wrapper around it, and a system that can decide, act and check its own results; how the three differ and relate.

03 **AI Agents and Agentic Systems**
How an agent takes in its situation, reasons about it, breaks a goal into steps, acts, and then observes the outcome before the next step. Everyday and engineering examples of agents in action.

04 **Prompting Fundamentals**
Giving clear instructions, context and examples, and asking for a specific output format.

Tools & tech stack

Python 3 Free-tier LLM API Ollama (local, optional) Google Colab

Before you arrive

Basic Python. A free LLM API key or Ollama set up per the pre-workshop guide. No paid subscription is required.

Why it matters on the car Engineering teams increasingly use agents to analyse logs, search documentation and automate routine checks. Understanding how they work is the first step to using them well.

SESSION6 AGENTICAI

Agentic AI Basics 2

Date	Tuesday, 3 Nov 2026	Time	5:10 PM 6:50 PM
Subject	Agentic AI	Presented by	**Arya, Josana**
Mini project	**Mini Project 6 (Agentic AI)**		

In this session, agents get tools and guardrails. You will see how an agent decides which tool to use, and build a simple agentic application safely and responsibly.

What you will learn

- Explain what tools are and how agents use them to reach external information and take actions
- Build a simple agentic AI application that uses tools to complete a task

- Identify engineering problems where an agent adds real value
- Apply basic safety checks, evaluate an agent's output, and follow responsible-AI practices

Topics covered

01 **Tools & Decision Making**
What a tool is (search, a calculator, a file reader, an API), how an agent decides to call one, and how results are fed back into its reasoning.

02 **Building Simple Agentic AI Applications**
Putting it together in Python: a goal, a few tools, the agent loop, and a working application you can run and extend.

03 **Engineering Applications of AI**
Analysing logs, searching documentation and automating routine checks on a vehicle project.

04 **Safety, Evaluation & Responsible AI**
Guardrails and step limits, keeping a human in the loop, checking whether an agent's answers are correct, and using AI fairly, transparently and responsibly.

Tools & tech stack

Python 3 LLM API / Ollama Google Colab

Before you arrive

Agentic AI 1 (Session 5) completed and a working API key or local model.

Why it matters on the car Tools, guardrails and honest evaluation are what turn a clever demo into a dependable engineering tool.

SESSION 7 · MACHINE LEARNING

Machine Learning: Supervised Learning, From the Basics

Date	Tuesday, 10 Nov 2026	Time	5:10 PM 6:50 PM
Subject	Machine Learning	Presented by	**Jeevan**
Mini project	**Mini Project 7 (ML)**		

From intuition to a working model: predict a continuous value from vehicle data and learn to judge whether the model can be trusted.

What you will learn

- Explain linear regression intuitively: fitting a line, loss and error
- Train a regression model in scikit-learn with a proper train/test split
- Judge a model with MAE and R , and recognise over- and under-fitting

Topics covered

01 **Linear Regression Demystified**
Predicting a continuous value, such as fuel or battery consumption, from throttle percentage. The best-fit line, mean squared error, and what the slope actually means.

02 **Training in scikit-learn**
Splitting data into train and test sets, fitting the model, and making predictions on unseen data.

03 **Evaluating a Model**
MAE, R , reading a residual plot, and spotting over-fitting before it fools you.

Tools & tech stack

Python 3 pandas scikit-learn Matplotlib Google Colab

Before you arrive

Session 4 notebook reviewed. Google account for Colab.

Why it matters on the car A regression model that has been tested honestly is the simplest useful prediction tool on a vehicle, and the baseline every fancier model is compared against.

SESSION 8 · ROS 2

ROS 2 and Software Engineering Fundamentals

Date	Friday, 13 Nov 2026	Time	5:10 PM 6:50 PM
Subject	ROS 2	Presented by	**Ratheeshwar**
Mini project	**Mini Project 8 (ROS)**	Project	**Project 2 is done after the workshop; details to be notified**

ROS 2 is the open-source framework used across research labs, industry robots and autonomous-vehicle teams. You will learn how it lets dozens of independent programs share data in real time, and drive your first simulated robot from the terminal.

What you will learn

- Explain what ROS 2 is, why robots use it, and how it differs from writing one giant program
- Describe the core primitives: nodes, topics, messages, and how they form a computation graph
- Use the essential CLI: ros2 run, node list, topic list / echo / hz / pub, interface show
- Visualise a running system with rqt_graph and explain every arrow in it
- Map the publisher-subscriber model onto real vehicle hardware

Topics covered

01 **Why ROS 2?**
Modularity, reuse and language independence (Python and C++). Where ROS 2 is used today, and why ROS 1 is no longer the starting point (ROS 1 Noetic reached end-of-life in May 2025).

02 **Core Primitives**
Nodes (individual programs), topics (named data channels) and messages (typed data packets such as std_msgs/Float32 or sensor_msgs/Image).

03 **The Publisher-Subscriber Model**
The onboard analogy: a wheel-speed sensor node publishes on /wheel_speed; the dashboard and the controller both subscribe, and neither needs to know the other exists.

04 **Essential ROS 2 Commands**
Running nodes, listing and echoing topics, measuring publish rates, publishing test messages by hand, and inspecting message types.

05 **Hands-on with turtlesim**
The classic ROS 2 simulator: drive it with teleop, then take over with your own velocity commands from the CLI.

06 **Software Engineering Fundamentals**
Software engineering practice for vehicle projects; the detailed outline will be shared by the presenter.

Tools & tech stack

ROS 2 Jazzy Jalisco Ubuntu 24.04 / WSL2 turtlesim rqt_graph

Before you arrive

ROS 2 installed as per the pre-workshop session. Comfort with basic terminal commands (cd, ls).

Why it matters on the car Autonomous vehicles are built as many small ROS nodes (camera driver, detector, planner, controller) talking over topics. After today you can read that system.

WEEKENDSESSION FUNDAMENTALS

Fundamentals of Automobiles and EVs

Date	Saturday, 14 Nov 2026	Time	To be notified
Subject	Fundamentals	Presented by	**To be notified**

The content outline, presenter and timing for this session will be shared by the workshop team.

WEEKENDSESSION FUNDAMENTALS

Fundamentals of Autonomous Vehicles

Date	Sunday, 15 Nov 2026	Time	To be notified
Subject	Fundamentals	Presented by	**To be notified**

The content outline, presenter and timing for this session will be shared by the workshop team.

4. Subject-wise Summary

Subject	Presented by	Sessions	Dates	Hours
System Design	Rithvin, Preethika	2	8 Oct, 13 Oct	3 h 20 min
Computer Vision	Mr. Mahavishnu V C	1	15 Oct	1 h 40 min
Machine Learning	Jeevan	2	22 Oct, 10 Nov	3 h 20 min
ROS 2	Ratheeshwar	1	13 Nov	1 h 40 min
Agentic AI	Arya, Josana	2	27 Oct, 3 Nov	3 h 20 min
Total		**8**	8 core sessions (Tuesday, Thursday and Friday 13 Nov)	**13 h 20 min**

The pre-workshop online session (6 Oct), the Saturday industry expert sessions, the fundamentals sessions (14 and 15 Nov), the guest sessions and the Sunday catch-ups are not counted in these hours.

5. Mini Projects and Projects

Every session conducted by the team has its own mini project. In addition, there are two larger projects. Briefs, team formation and deadlines will be shared in due course.

Mini projects

Mini project	Session	Date
Mini Project 1 (System Design)	Session 1 System Design I: Subsystem Thinking	Thursday, 8 Oct 2026
Mini Project 2 (System Design)	Session 2 System Design II: Block Diagramming	Tuesday, 13 Oct 2026
Mini Project 3 (Computer Vision)	Session 3 Computer Vision: Teaching the Car to See	Thursday, 15 Oct 2026
Mini Project 4 (ML)	Session 4 Introduction to ML and Unsupervised Training	Thursday, 22 Oct 2026
Mini Project 5 (Agentic AI)	Session 5 Introduction to Agentic AI 1	Tuesday, 27 Oct 2026
Mini Project 6 (Agentic AI)	Session 6 Agentic AI Basics 2	Tuesday, 3 Nov 2026
Mini Project 7 (ML)	Session 7 Machine Learning: Supervised Learning, From the Basics	Tuesday, 10 Nov 2026
Mini Project 8 (ROS)	Session 8 ROS 2 and Software Engineering Fundamentals	Friday, 13 Nov 2026

Projects

Project	When	Builds on
Project 1	Assigned after the first Computer Vision session (Session 3 Thu 15 Oct)	System Design and Computer Vision
Project 2	Done after the workshop (after Session 8 Fri 13 Nov, and the 14 15 Nov fundamentals sessions)	The full workshop: System Design, Computer Vision, Machine Learning, Agentic AI and ROS 2

Student Timetable

Software & Perception Workshop 2026 Team ASTERIX aBAJA SAEINDIA

Core sessions: 5:10 PM 6:50 PM **Saturdays: industry expert session (half day, timing to be notified)** **Sundays: weekly catch-up at 6:00 PM** **14 15 Nov: fundamentals sessions**

Week	Monday	Tuesday	Wednesday	Thursday	Friday	Saturday	Sunday
Week 1 *5 Oct 11 Oct*	**5 Oct**	**6** PRE PRE-WORKSHOP SESSION **Pre-Workshop** *Online session* *by all team members*	**7**	**8** S1 SYSTEM DESIGN **System Design I** *by Rithvin, Preethika* *Mini Project 1*	**9**	**10** INDUSTRY EXPERT **Industry Session** *Half day* *Timing to be notified*	**11** CATCH-UP **Weekly Catch-up** *6:00 PM*
Week 2 *12 Oct 18 Oct*	**12**	**13** S2 SYSTEM DESIGN **System Design II** *by Rithvin, Preethika* *Mini Project 2*	**14**	**15** S3 COMPUTER VISION **Computer Vision** *by Mr. Mahavishnu V C* *Mini Project 3* ***Project 1 assigned***	**16**	**17** **No session** *Vijayadashami*	**18** CATCH-UP **Weekly Catch-up** *6:00 PM*
Week 3 *19 Oct 25 Oct*	**19**	**20** **No session** *Vijayadashami*	**21**	**22** S4 MACHINE LEARNING **Intro to ML & Unsupervised** *by Jeevan* *Mini Project 4*	**23**	**24** INDUSTRY EXPERT **Automotive Functional Safety** *Half day* *Timing to be notified*	**25** CATCH-UP **Weekly Catch-up** *6:00 PM*
Week 4 *26 Oct 1 Nov*	**26**	**27** S5 AGENTIC AI **Intro to Agentic AI 1** *by Arya, Josana* *Mini Project 5*	**28**	**29**	**30**	**31** INDUSTRY EXPERT **Industry Session** *Half day* *Timing to be notified*	**1 Nov** CATCH-UP **Weekly Catch-up** *6:00 PM*
Week 5 *2 Nov 8 Nov*	**2**	**3** S6 AGENTIC AI **Agentic AI Basics 2** *by Arya, Josana* *Mini Project 6*	**4**	**5**	**6**	**7** INDUSTRY EXPERT **Industry Session** *Half day* *Timing to be notified*	**8** CATCH-UP **Weekly Catch-up** *6:00 PM*
Week 6 *9 Nov 15 Nov*	**9**	**10** S7 MACHINE LEARNING **Supervised Learning** *by Jeevan* *Mini Project 7*	**11**	**12**	**13** S8 ROS 2 **ROS 2 & Software Eng.** *by Ratheeshwar* *Mini Project 8* ***Project 2 after the workshop***	**14** FUNDAMENTALS **Automobiles and EVs** *Timing to be notified*	**15** FUNDAMENTALS **Autonomous Vehicles** *Timing to be notified*

Pre-Workshop Session	System Design	Computer Vision	Machine Learning	ROS	Agentic AI	Industry Expert Session	Fundamentals	Weekly Catch-up	Guest Session	No Session (Holiday)

Each core session has its own mini project. Attendance is monitored; certificates only for participants who meet the attendance requirement. Project 1 is assigned after the Computer Vision session (Thu 15 Oct); Project 2 is done after the workshop. The industry expert session on Vehicle Dynamics and Architecture by Mr. Senthil Vel M (VIT) and Mr. Vimal Kumar (MEI) falls on one of the Saturdays shown; the date will be notified. Guest sessions by seniors and alumni are not shown; their dates will be communicated in due course.

6. Industry Expert Sessions, Guest Sessions and Support

Industry expert sessions (Saturdays)

An industry expert session is held on every Saturday of the workshop except 17 October (Vijayadashami) and 14 November (fundamentals session): 10, 24 and 31 October and 7 November 2026. Each runs for half a day; timings will be informed later.

CONFIRMEDSESSION

Automotive Functional Safety

Saturday, 24 Oct 2026 half day, timing to be notified speaker to be notified

CONFIRMEDSESSION

Vehicle Dynamics and Architecture

Mr. Senthil Vel M (VIT) and **Mr. Vimal Kumar** (MEI) Saturday, date and timing to be notified

The remaining Saturdays will also be handled by industry experts. Speakers and topics will be informed later.

Fundamentals sessions (weekend of 14 15 Nov)

The workshop closes with two fundamentals sessions: Fundamentals of Automobiles and EVs on Saturday, 14 November 2026, and Fundamentals of Autonomous Vehicles on Sunday, 15 November 2026. Timings and presenters will be informed later.

Guest sessions by seniors and alumni

Arranged by the team and conducted by senior students and alumni of the college. Dates will be communicated in due course. No project is attached to these sessions.

Guest Session I: Powertrain and Embedded/Electrical Fundamentals ***Where code meets current***

Software eventually has to switch a relay, read a sensor or spin a motor. This session covers the electrical and embedded foundations every autonomy engineer needs to talk to the hardware team.

- **Basic electrical concepts.** Voltage, current, resistance, power (Ohm's law), series and parallel circuits, and reading a simple schematic.
- **Digital and analog signals.** Logic levels, analog-to-digital conversion, sampling, resolution and PWM.
- **Microcontrollers.** Arduino and ESP32 class boards: CPU, memory, pins, clocks, and how they work alongside an onboard computer.
- **Embedded programming fundamentals.** The setup/loop structure, timing, reading sensors and writing outputs.
- **GPIO and peripheral interfaces.** GPIO, UART, I C, SPI and a first look at CAN bus in vehicles.

Tools: Wokwi simulator Arduino IDE Arduino / ESP32

Guest Session II: Mechanical Fundamentals ***Know the machine you're coding for***

Great autonomy software respects the physics of the vehicle underneath it. This session walks through the anatomy of an off-road ATV so your code makes sense on real terrain.

- **Vehicle architecture.** Overall layout of an off-road ATV and how subsystems are packaged.
- **Chassis and components.** Frame design, materials and mounting points.
- **Steering and suspension.** Steering geometry, suspension travel, and the effect on camera and IMU readings.
- **Braking and drivetrain.** Brake circuits, transmission and power delivery to the wheels.
- **Sensor mounting.** Field of view, vibration isolation, weather and impact protection, and calibration.
- **Basics of vehicle dynamics.** Weight transfer, traction limits, understeer/oversteer and turning radius.

Tools: Vehicle outline sheets Pen & paper

Attendance and certificates

Attendance will be monitored for every session. Certificates will be given only to participants who meet the attendance requirement.

Learning support for every participant

- **Weekly catch-up sessions (every Sunday, 6:00 PM).** Held on 11, 18 and 25 October and 1 and 8 November 2026 so that no participant is left behind. Participants who miss a class can revisit the missed content, clear doubts and keep pace with the full syllabus.
- **Handbooks, guides and study material.** All material required to complete the course is provided to every participant, and stays a useful reference for future academic and project work.
- **Extra time where needed.** If a topic needs more time than its session allows, supplementary online sessions or other arrangements will be made so the planned content is covered.

Participants are encouraged to direct any questions or doubts to the workshop team at any point, and to make full use of the sessions, resources and hands-on opportunities provided. Difficulties and suggestions for improving the workshop are welcome.

** The syllabus, content covered, schedule and other workshop details are subject to change with or without prior notice.*